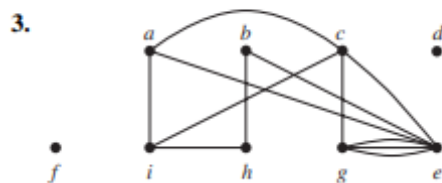
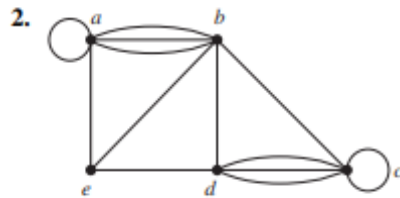
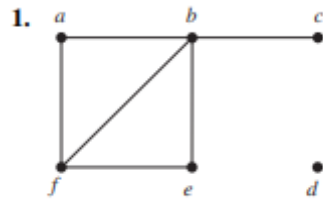


F.Y.B.Sc Data Science Sem II

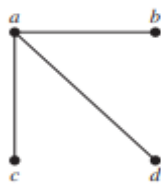
GRAPH THEORY PRACTICALS (THEORY)

Practical no. 1: Introduction to graphs-I

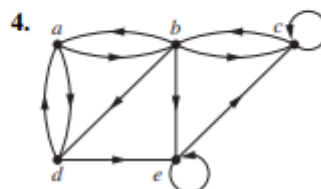
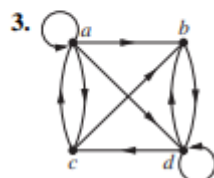
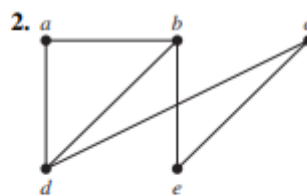
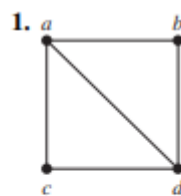
- Find the number of vertices, the number of edges, and the degree of each vertex in the given undirected graph.



- Draw the graphs: a) K_7 b) $K_{1,8}$ c) $K_{4,4}$ d) C_7 e) W_7
- Draw all the subgraphs of the following graph:

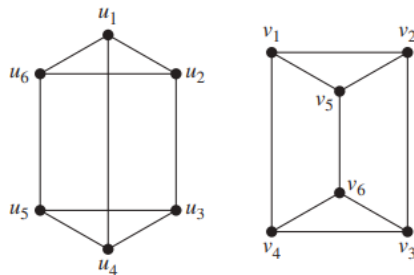
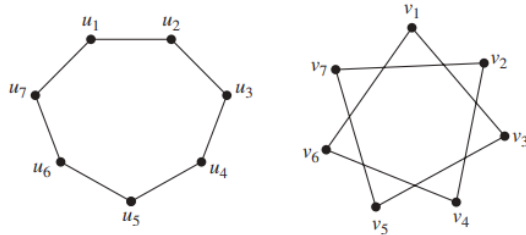
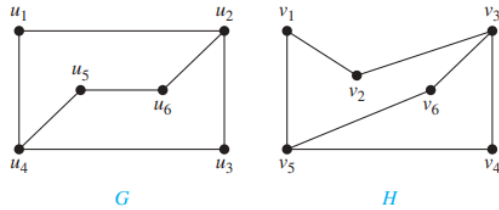


- Represent each of these graphs with an adjacency matrix. a) K_7 b) C_7
- Represent the following graphs with an adjacency matrix.

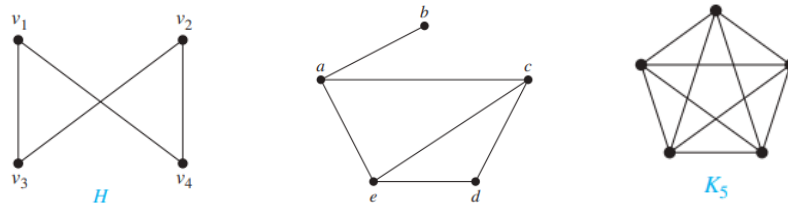


Practical no. 2: Introduction to graphs-II

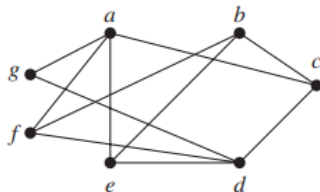
1. Determine whether the graphs shown below are isomorphic or not.



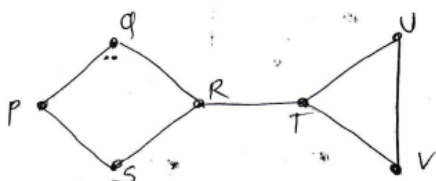
2. Find the complement of the following graphs:



3. Determine whether the following graph is a bipartite graph. Justify.



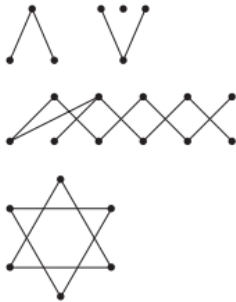
4. Give an example of a graph which has:
- A Hamilton circuit but not an Euler circuit.
 - An Euler circuit as well as a Hamilton circuit.
5. For the following graph G, answer the following questions:



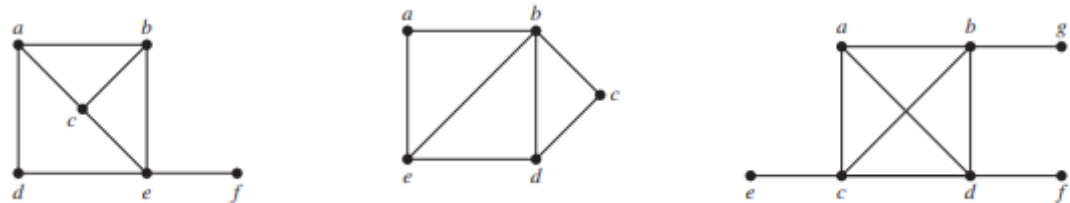
- List all the cut vertices in G.
- List any two distinct paths from vertex P to vertex U in G.
- List all the cut edges in G.
- List all the cycles in G.

Practical no. 3: Connected Graphs

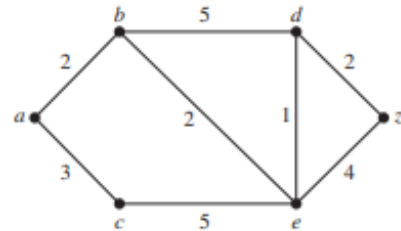
1. Determine whether the given 3 graphs are connected:



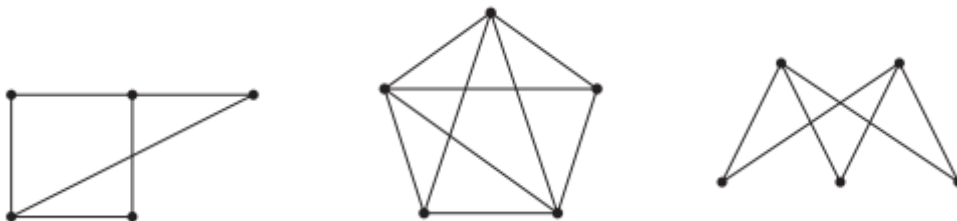
2. Determine whether the given graph has a Hamilton circuit. If it does, find such a circuit.



3. Find the shortest path from a to z in the graph below:

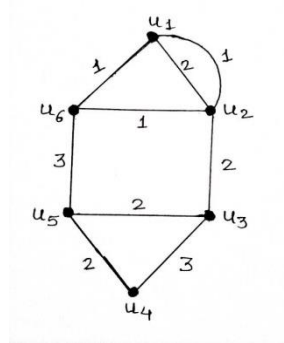
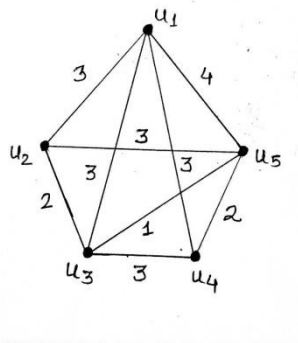
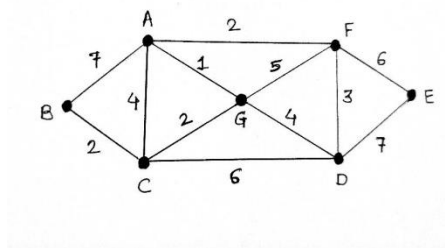


4. Draw the following planar graphs without any crossings:

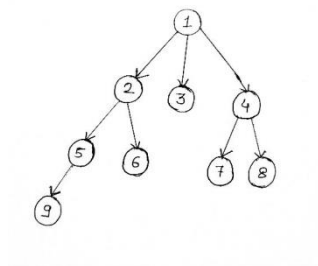
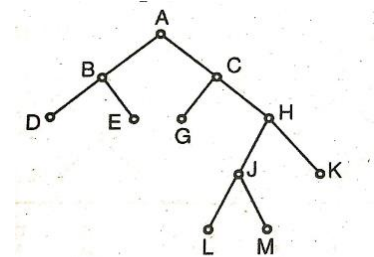


Practical no. 4: Trees

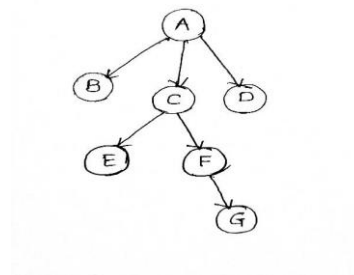
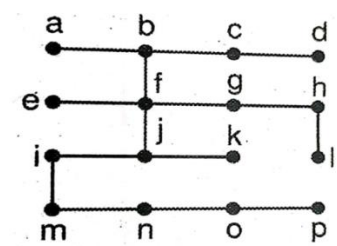
1. Find the shortest spanning tree of the following graphs :



2. Find the preorder and postorder traversal of the following binary trees:



3. Find the BFS of the following Trees:



4. Find the DFS of the following Trees:

